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Econ 140 Final Project

Finding stable, well-paying employment is essential for quality of life, but social and structural barriers often limit access. Therefore, for our final project, we ask: Does the economic return to physical attractiveness differ between Black and White female workers? Using the 'beauty' dataset from the Wooldridge repository, my team aims to better understand the relationship of the outcome variable, log wage, and the explanatory variable, physical attractiveness (rank 1-5). We would like to focus on how this relationship differs by race, given the importance of wages in shaping everyday life. We hypothesize that returns to physical attractiveness does differ between Black & White female workers and suggests that Black women will face a larger beauty penalty from systematic racial and gender biases, as well as labor market discrimination.

Before running any regressions, we filtered the 'beauty' dataset to include only females. We then examined the physical attractiveness variable and found it to be a subjective measure, so we checked for balance. The average attractiveness rating was 3.21 for White women and 3.12 for Black women, only showing a 2.8% difference. Given this small gap, we treated the dataset as balanced and continued with our analysis. As a preliminary action, we plotted an overlaid histogram demonstrating the overall spread of wages between Black and White women, which portrayed White women wages are on average higher than Black Women.

Our first regression of ***lm(formula = lwage ~ looks + educ + married + goodhlth, data = female_beauty_data)*** was then created to explore the general return to attractiveness without accounting for race. To address potential omitted variable bias, we controlled for education level, marital status, and health, as these factors often influence employment outcomes. This regression output reveals that a one unit increase in *looks* on average, for both White and Black women, leads to a 3.4% increase in *wage*, but is not statistically significant (T-value: 1.06) which we found surprising as we would expect this increase to be historically accurate. Our second regression analysis of ***lm(formula = lwage ~ looks + educ + married + goodhlth + blackxlooks, data = female_beauty_data)*** was created to explore the factor of race on wage, by including an interaction term between Black women and looks on log wage. In this regression, the coefficient for looks captures the return to attractiveness for White women, while the interaction term *blackxlooks* measures how the return to beauty differs for Black women. The results of this regression demonstrates a one unit increase in *looks* for Black women leads to a 4.43% increase in *wage* and is statistically significant (T-value: 2.38). It also demonstrated a one unit increase in *Education* has a 7.9% percent increase in *wage* and is statistically significant (T-value: 8.29).

These regression results are actually opposite of our hypothesis, where Black women face a higher economic return than White women. One possible explanation is that Black women face more scrutiny based on appearance, so being perceived as highly attractive may yield a greater wage benefit compared to White women, who face less bias and thus experience more moderate returns.While our results suggest a positive return to attractiveness for Black women, we cannot make causal claims. The dataset is limited by its small and unbalanced sample (50 Black women & 386 White women) and outdated (from 1994). Additionally, physical attractiveness is a subjective measure, which introduces a measurement error into the analysis. To better understand this relationship and establish causality, a randomized controlled trial with a larger and more balanced sample would be ideal.

Our First Regression Results

```
Call:
lm(formula = lwage ~ looks + educ + married + goodhlth, data = female_beauty
_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.28573 -0.31116  0.04698  0.30769  3.01283 

Coefficients:
             Estimate Std. Error t value            Pr(>|t|)    
(Intercept)  0.211115   0.165918   1.272               0.204    
looks        0.034753   0.032800   1.060               0.290    
educ         0.077044   0.009292   8.291 0.00000000000000144 ***
married     -0.043925   0.046761  -0.939               0.348    
goodhlth     0.032507   0.086081   0.378               0.706    
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.485 on 431 degrees of freedom
Multiple R-squared:  0.1523,	Adjusted R-squared:  0.1444 
F-statistic: 19.36 on 4 and 431 DF,  p-value: 0.00000000000001169
```

Our Second Regression Results

```
Call:
lm(formula = lwage ~ looks + educ + married + goodhlth + blackxlooks, 
    data = female_beauty_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.27281 -0.32371  0.04108  0.31739  2.81280 

Coefficients:
             Estimate Std. Error t value            Pr(>|t|)    
(Intercept)  0.164938   0.166159   0.993              0.3214    
looks        0.030573   0.032670   0.936              0.3499    
educ         0.079609   0.009304   8.556 <0.0000000000000002 ***
married     -0.035528   0.046642  -0.762              0.4466    
goodhlth     0.036566   0.085634   0.427              0.6696    
blackxlooks  0.054281   0.022779   2.383              0.0176 *  
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4824 on 430 degrees of freedom
Multiple R-squared:  0.1633,	Adjusted R-squared:  0.1536 
F-statistic: 16.79 on 5 and 430 DF,  p-value: 0.000000000000003659
```
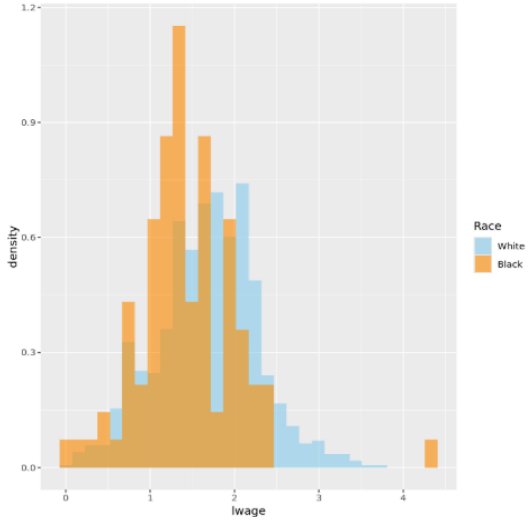
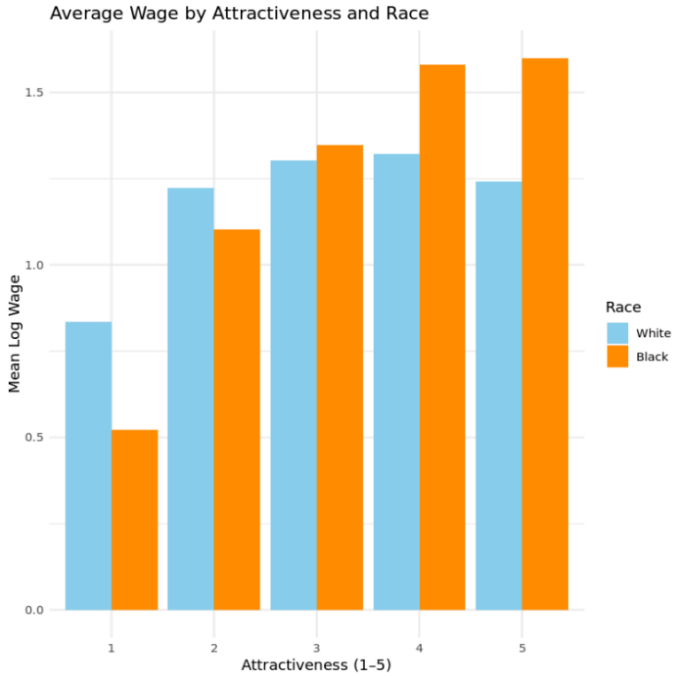

Overlaid histogram demonstrating that overall, on average, white women's spread of wages is shifted right slightly, demonstrating higher wages on average.



Graph picturing that very attractive Black women (ranked 4-5) see much higher returns to beauty than White women with the same rank.